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2. In 1992 scientists first detected a planet orbiting a star outside our Solar System. Hundreds of these exoplanets have been found since. **Table 1** shows the stars that have been discovered with 6 or more exoplanets in orbit, along with information about our Sun.

Table 1

Name of star	Distance of star from Earth (l-y)	Mass of star (compared to the Sun)	Temperature of star (K)	Number of planets in orbit
Sun	0.00002	1.0	5770	8
HR 8832	21	0.8	4700	7
HD 10180	127	1.1	5910	7
Kepler 90	2500	1.1	5930	7
HD 40307	42	752.0	4980	6
HD 127	206	0.7	870	6
Kepler 20	950	0.9	5470	6
Kepler 11	2000	1.0	5680	6

(a) (i) Tick (✓) the **two** correct statements below. [2]

Four stars have the same mass. ☐

Light travelling from Kepler 90 takes the longest time to reach Earth. ☐

The range of star temperatures is 5060 K. ☐

Hotter stars have more planets in orbit around them. ☐

Star HR 8832 is double the distance of star HD 40307 away from Earth. ☐

(ii) State **two** reasons, using evidence from **Table 1**, why scientists predict that the star Kepler 11 is most likely to have a Solar System similar to ours. [2]

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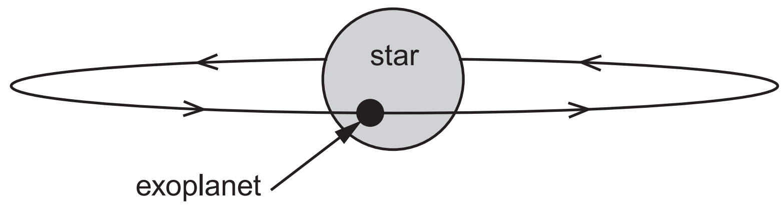
(b) Exoplanets can't be detected directly because they don't give out their own light. One method of detection is to look for a slight dip in the light intensity from the star. This happens when the exoplanet passes in front of the star during its orbit. This is called a transit.



(i) Why does the detected light intensity decrease during a transit? [1]

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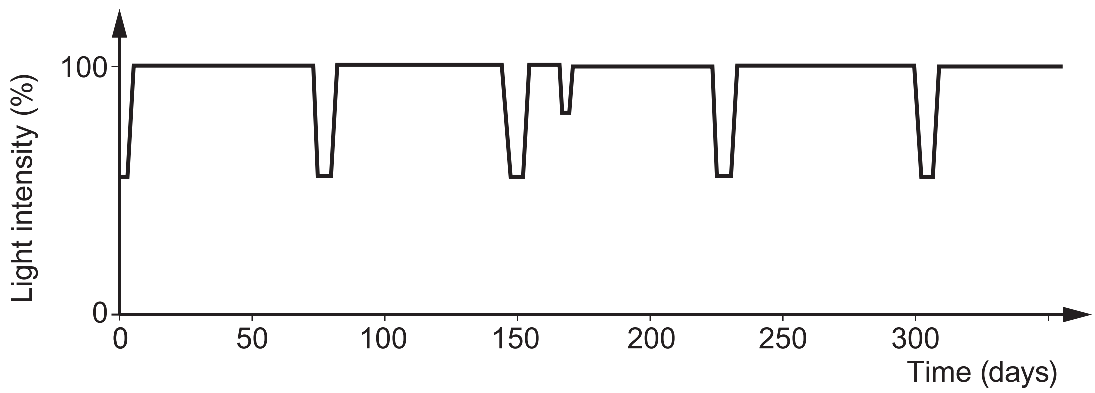
(ii) A pupil finds a graph on the internet that shows the measured light intensity detected from a star against time.



I. Calculate the orbit time, in days, of the exoplanet. [2]

Orbit time = ... days

II. The pupil correctly suggests that there may be two exoplanets in orbit around this star. What evidence is there to back up this claim? [1]

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III. During a transit some of the light from the star passes through the exoplanet's atmosphere. An absorption spectrum is produced. Explain how the absorption spectrum arises and is used to identify gases in its atmosphere. [2]

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(c) **Table 2** shows data collected for all the exoplanets in orbit around the star called Kepler 11. They are listed in order of increasing distance.

Table 2

Exoplanet	Mass (compared to Earth)	Time for one orbit (days)	Temperature (K)	Orbit radius (AU)
b	1.9	10.3	871	0.09
c	2.9	13.0	807	0.11
d	7.3	22.7	659	0.16
e	8.0	32.0	596	0.20
f	2.0	46.7	525	....
g	25.0	118.4	386	0.47

(i) Compare the **trend** in temperature with orbit radius of the planets around Kepler 11 to the planets in our Solar System where Venus is the hottest. [2]

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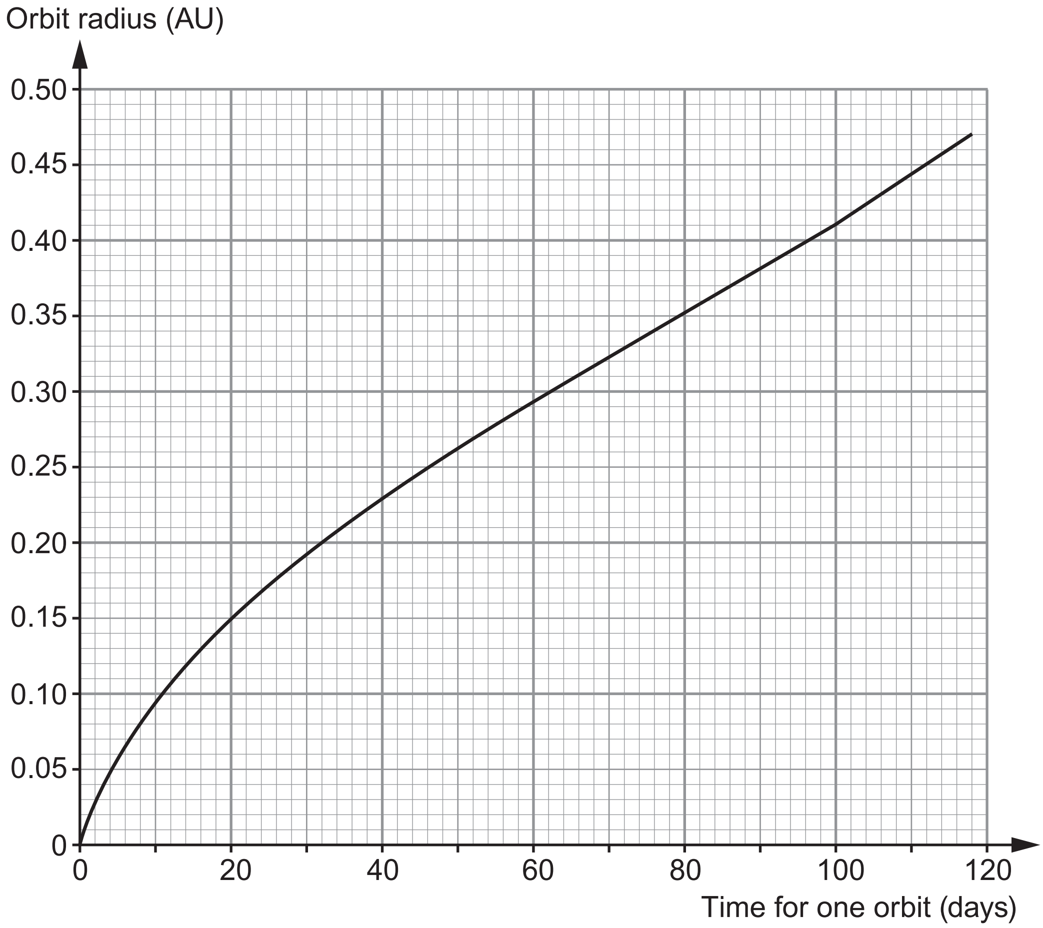
(ii) Some of the data from the table is used to plot a graph.



The orbit time of **exoplanet f** was found using the transit method. Its orbit time was measured as 46.7 days. Use the graph to **complete Table 2** with an estimate of a value for the missing orbit radius. [1]

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